

Problem D. Longest Common Subsequence

Time Limit 1000 ms

Mem Limit 524288 kB

Given two arrays of integers, find their longest common subsequence.

A subsequence is a sequence of array elements from left to right that can contain gaps. A common subsequence is a subsequence that appears in both arrays.

Input

The first line has two integers n and m : the sizes of the arrays.

The second line has n integers $a_1, a_2, \dots, a_n$: the contents of the first array.

The third line has m integers $b_1, b_2, \dots, b_m$: the contents of the second array.

Output

First print the length of the longest common subsequence.

After that, print an example of such a sequence. If there are several solutions, you can print any of them.

Constraints

- $1 \leq n, m \leq 1000$
- $1 \leq a_i, b_i \leq 10^9$

Example

Input	Output
8 6 3 1 3 2 7 4 8 2 6 5 1 2 3 4	3 1 2 4